

Muhan Li

778-680-8519 | dev@muhanli.ca | linkedin.com/in/muhan-li | github.com/muhan-alan-li

EDUCATION

University of British Columbia

BASc. Computer Engineering – GPA: 3.8

Sep 2020 - May 2026

Vancouver, BC

EXPERIENCE

Software Engineer

Apera AI

Sep 2025 - Apr 2026

Vancouver, BC

- Shipped self-serve asset deployment with live status tracking (REST API, React/TypeScript), replacing a manual support-ticket workflow and cutting deployment time from ~12hrs to ~5min
- Extended Forge's Python DAG job scheduler with a new node type for self-serve asset deployment, using Docker Compose for isolated execution and scoped secrets, validated with Pytest test suite
- Refactored object-sharing logic to use DynamoDB transactional writes, preventing dangling object references during failures
- Reworked Redux state management and validation to catch invalid object references before downstream workflow errors
- Built real-time collision detection using spatial hashing in Three.js, maintaining 60+ FPS through React memoization

Software Developer

Natural Resources Canada

May 2025 - Aug 2025

Vancouver, BC

- Ported the ANSE simulation engine from Windows-only .NET to cross-platform, enabling Linux and macOS support
- Replaced legacy data transformation library with a C# tool for bidirectional conversion across inconsistent Excel/text formats
- Configured GitHub Actions CI/CD pipelines, containerizing builds and deployment for the ANSE engine across environments
- Built data dashboard with parameterized SQLite queries in Electron, reducing manual data retrieval time by ~50%

Software Developer

UBC Cloud Innovation Center

Jan 2023 - Aug 2023

Vancouver, BC

- Built a serverless video pipeline with AWS Rekognition and Lambda for video capture and facial anonymization
- Shipped an anonymous drug safety testing platform with React and AWS, using JWT and RBAC to protect user data
- Delivered projects for two non-profit clients, translating evolving requirements into deployable software

Software Test Engineer

Sierra Wireless Inc. (now Semtech)

May 2022 - Dec 2022

Richmond, BC

- Automated manual regression testing with internal libraries, owning test case design and execution end-to-end
- Diagnosed modem crashes with firmware and hardware engineers, driving defects from discovery through resolution
- Triaged customer reported issues, escalating critical bugs to firmware team with reproducible test cases

TECHNICAL PROJECTS

Holdr – Artist Time Vault | Next.js, Nest.js, GraphQL, PostgreSQL

Sep 2024 - Apr 2025

- Designed Nest.js backend with PostgreSQL and S3 for secure file storage and streaming
- Implemented hierarchical file organization, soft deletion, and metadata tagging through GraphQL APIs
- Enabled large file up/downloads with presigned URLs, reducing client-server payload size and load times by 70%

Find A* | Node.js, MySQL, Prisma, Android

Sep 2023 - Nov 2023

- Implemented a friendship system with MySQL and Prisma, supporting requests, adds, and removes via REST APIs
- Developed weighted review algorithm with peer feedback + time-based-decay to prioritize recent, relevant reviews

TECHNICAL SKILLS

Languages: TypeScript, Python, C++, C#, JavaScript, HTML/CSS, SQL

Frameworks: React, Node.js, Nest.js, GraphQL, REST APIs, .NET, Flask, Three.js, NoSQL

Tools: Agile, Claude Code, Git, Docker, AWS, Jira, Ubuntu, Terraform, Cloudwatch